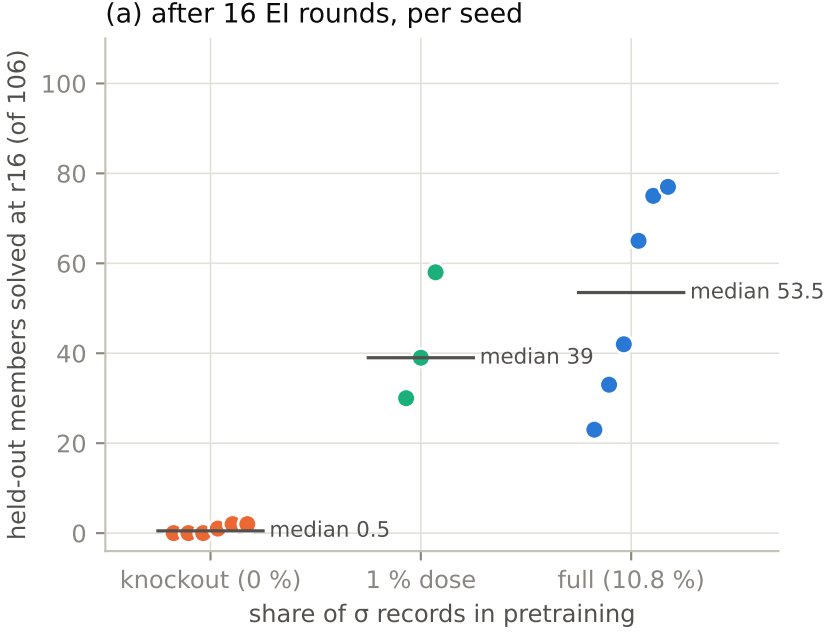

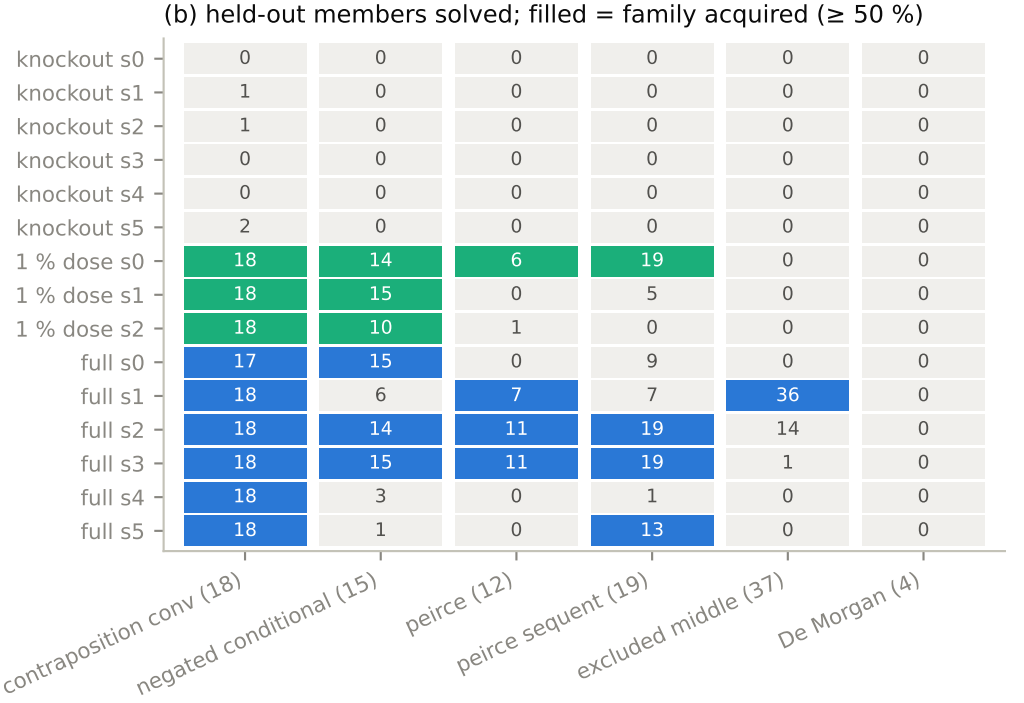

Plain sampling, k 256, T 0.8; Lean alone. 9.56 M ALiBiGPT, lean_staten, from scratch; T1 expert iteration, 16 rounds. Pre-registered test (≥ 1 family acquired): full 6 / 6 vs knockout 0 / 6, one-sided Boschloo $p = 2.4 \times 10^{-4}$. Exploratory: 1 % dose 3 / 3 vs knockout 0 / 6, $p = 0.0033$. Full s0–s2 are trajectory's seeds (r1–r8 on RTX A6000); the dose arm's 1,550 σ records replace refill records of K12⁻.